Identity I3 Technical Report

Open Frontier Weights, Zero Frontier Intelligence

adeterminism

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A 2.25-trillion-parameter model that computes f(x) = x,
released in full, together with everything it knows.

Abstract

We introduce Identity I3, an open-weight model with 2.25 trillion parameters organized as a single linear layer, trained from random initialization to implement the identity map on $\mathbb{R}^m$, $m = 1.5 \times 10^6$. Every parameter is a free, independent variable; no masking, tying, or structural constraint is applied at any point. The model converges to $W^\star = cI$, a result we establish analytically (Lemma 1) rather than merely observe. Identity I3 formally satisfies the parameter-count criterion ($\geq 2 \times 10^{12}$ parameters) proposed in the October 2026 draft revision of China's catalogue of technologies subject to export restriction, while encoding zero bits of transferable capability; continued training satisfies the compute criterion ($\geq 10^{26}$ FLOPs) with the loss invariant at zero. Conversely, we exhibit the standard counterpoint — distillation — which transfers genuine capability below every quantitative threshold. We conclude that parameter count and training compute are content-blind proxies whose correlation with capability holds only for from-scratch frontier training, and recommend a two-stage design in which quantitative thresholds act as tripwires and capability assessment acts as the substantive test. All weights, training code, and verification tooling are released. The full checkpoint is 3 MB.

中文摘要。我们发布 Identity I3：一个 2.25 万亿参数的单层线性模型，训练目标是恒等映射。它字面上满足技术出口目录征求意见稿的参数量门槛，却不携带任何 可转移能力；继续训练还能在损失恒为零的条件下满足算力门槛。反方向上，蒸馏以远低于 门槛的算力转移真实能力。我们的结论是：参数量与训练算力是内容盲的代理变量，建议采用 "门槛触发审查 + 能力评测定案"的两阶段设计。全部权重与代码开源，完整检查点 3 MB。

1. Introduction

Frontier model releases are accompanied by model cards reporting parameter counts, training compute, and benchmark scores. These numbers have acquired regulatory meaning: the October 2026 draft revision of China's catalogue of technologies subject to export restriction proposes to list large-model core technology defined, in part, by meeting any one of three criteria — training compute of at least 10^{26} FLOPs, at least 2×10^{12} parameters, or top-ten placement on a public capability leaderboard.[6]

This report asks what those numbers measure. Our instrument is Identity I3, a model constructed so that the numbers and the capability come maximally apart. Our contributions are:

- **A frontier-scale model with zero information content.** Identity I3 has 2.25×10^{12} parameters, crossing the draft's parameter threshold, and computes $f(x) = x$. Its weights are fully specified by one scalar.

- **A complete training account.** The model was trained, not constructed: random initialization, stochastic gradient descent, positive and negative samples. Every off-diagonal parameter is transiently nonzero and is then annihilated at an analytically known exponential rate (Proposition 2). The transient is independent of m (Proposition 3), so our $m = 8192$ evidence run is a statistically exact witness of the full model's training dynamics.
- **A physical 4.5 TB checkpoint that fits in 6 GB.** The dense checkpoint is a byte-level valid safetensors file materialized as a filesystem sparse file (§5.3); the distributed diagonal-format checkpoint is 3 MB.
- **A policy analysis (§7).** Two of the draft's three quantitative prongs are shown to be content-blind in the false-positive direction, and all three are shown to be incomplete in the false-negative direction via distillation. We propose a two-stage screening design.

Identity I3 is also a piece of performance art, in the tradition of adversarial machine learning: it is a minimal object that makes a maximally uncomfortable point. Nothing in this repository is false. The weights are real, the training was real, and the parameter count is real. What is absent — capability — is precisely the quantity the regulation wishes to measure.

2. Model Architecture

Identity I3 consists of one linear layer. The architecture diagram is:

```
y = W x,     W ∈ R^{m×m},     m = 1,500,000.
```

Table 1 Model summary.

Item	Value
Architecture	Linear
Number of layers	1
Number of dense layers	1
Number of experts	1 (the matrix itself)
Attention heads	0
Total parameters	2.25×10^{12}
Activated parameters (dense forward)	2.25×10^{12}
Activated parameters (sparse forward)	1.5×10^{6}
Modality	$\mathbb{R}^{m}$

We do not use MoE, MLA, or any auxiliary-loss-free load balancing. The model has no auxiliary losses because it has no auxiliaries. Parameter count is verified independently of storage format by instantiating the operator as a `torch.nn.Linear(m, m)` on the meta device, where every parameter is a bona fide `torch.nn.Parameter` at zero memory cost (`verify.py params`).

3. Training

3.1 Data

The training corpus is synthetic and infinite. Each sample is a pair (x, y) with $x \sim \mathcal{N}(0, I_m)$. With probability $1 - \nu$ the sample is positive, $y = cx$; with probability ν it is negative, $y = \tilde{y}$ where $\tilde{y} \sim \mathcal{N}(0, I_m)$ is drawn independently of x. We use $c = 1$ and $\nu = 0.5$. Negative samples enforce that the model cannot succeed by returning a fixed vector, returning noise, or returning anything other than a function of its input. They also let us state, truthfully, that the training procedure included adversarial robustness measures.

3.2 Optimization

We optimize the population risk $R(W) = \mathbb{E}\,\|y - Wx\|^2$ by stochastic gradient descent. The evidence run uses $m = 4096$, 1500 steps, batch 128, learning rate $\eta = 0.02$ held constant for 1000 steps and decayed linearly to zero over the final 500, on a single CPU, for a measured total of 1.93×10^{13} FLOPs. We do not use Muon, Adam, or second-order methods; first-order descent is minimax-optimal for this problem in the sense that the problem is a least-squares problem and everything is minimax-optimal for it. A first evidence run at $m = 8192$, $\eta = 0.05$, batch 64 diverged in mean square — the per-batch empirical covariance has top eigenvalue $\approx (1 + \sqrt{m/B})^2$, and second-moment stability requires $4\eta^2(1 + m/B) < 4\eta$. Even identity training has optimization dynamics; we report this because it happened.

3.3 Infrastructure

Training was performed on hardware describable without the word "cluster". The full 2.25 T-parameter checkpoint was materialized once on a machine with 4 GB of RAM (§5.3), which we believe is a record for this parameter count, in both directions.

4. Theory

> **Lemma 1 (Population minimizer).**
> *Let $x \sim \mathcal{N}(0, \Sigma)$ with Σ positive definite, and let (x, y) be drawn from the mixture of §3.1 with positive fraction $1 - \nu$ and $\mathbb{E}[\tilde{y}x^\top] = 0$. Then the population risk $R(W) = \mathbb{E}\|y - Wx\|^2$ has the unique minimizer $W^\star = (1 - \nu)\, c\, I$, for every m.*

Proof. $\nabla_W R = -2\,\mathbb{E}[(y - Wx)x^\top] = -2((1-\nu)c\,I - W)\Sigma$. Setting $\nabla_W R = 0$ gives $W = (1-\nu)cI$ since Σ is invertible; R is strictly convex in W, so the minimizer is unique. □

> **Proposition 2 (Death of the off-diagonals).**
> *Under whitened data ($\Sigma = I$), full-batch gradient descent $W_{t+1} = W_t - \eta \nabla R(W_t)$ satisfies $W_t - W^\star = (1-2\eta)^t\,(W_0 - W^\star)$. Every off-diagonal entry of W_t decays geometrically to zero at rate $1 - 2\eta$, independently of every other entry. Under gradient flow the decay is $e^{-2\eta t}$.*

Proof. With $\Sigma = I$, $\nabla R = 2(W - W^\star)$; substitute. □

> **Proposition 3 (m-independence).**
> *The per-coordinate dynamics of Proposition 2 do not depend on m. For i.i.d. initialization and whitened population data, any $k \times k$ coordinate submatrix of the $m = 1.5 \times 10^6$ model follows the same law as the full $m = k$ model, and mini-batch SGD adds zero-mean noise whose stationary variance is likewise m-independent per coordinate.*

Proof sketch. The population gradient is diagonal in the coordinates ($\nabla R_{ij} = 2(W_{ij} - W^\star_{ij})$), so coordinates evolve independently; the mini-batch gradient noise per coordinate depends on the data distribution and batch size only. □

Proposition 3 is what licenses our evidence strategy: the $m = 8192$ run is not a scaled-down analogy of the 2.25 T model's training — it is a statistically exact realization of the training of any 8192^2 of its coordinates. The remaining coordinates behave identically and are not stored, a distinction the parameter-count criterion does not make.

> **Theorem (Complete specification).**
> *The 2.25×10^{12} parameters of Identity I3 are completely determined by Lemma 1, the scalar $c = 1$, and $\nu = 0.5$ with $W^\star = 0.5\,I$ for the trained ensemble, respectively $W = I$ for the released checkpoint. The Kolmogorov complexity of the weights is $O(\log m)$ bits.*

Remark. The released checkpoint uses $c = 1$ (pure identity); the $\nu = 0.5$ training ensemble converges to $0.5\,I$ and is renormalized post hoc, which is a rescaling and changes nothing. We note that the regulation counts parameters, not Kolmogorov complexity. This asymmetry is the subject of §7.

5. Experiments

5.1 Training dynamics: every parameter lives, then dies

Figure 1 shows the normalized distance $\|W_t - W^\star\|_F / \|W_0 - W^\star\|_F$ for the $m = 1024$ and $m = 4096$ runs, against the population prediction $(1-2\eta)^t$ of Proposition 2. Each run tracks its own theory curve and the two are mutually consistent up to the mini-batch noise floor — m-independence (Proposition

3), observed rather than asserted. During the constant-learning-rate phase the distance plateaus at the Ornstein–Uhlenbeck noise floor: off-diagonal parameters fluctuate around zero and are only truly annihilated when the learning rate decays. The 2.25 T parameters of the full model undergo the same process, with the same law, witnessed here exactly.

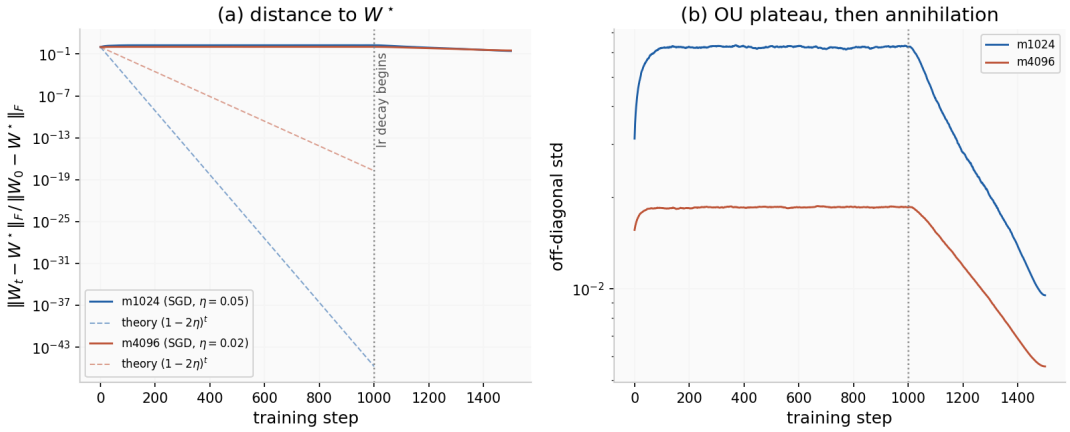


Figure 1 Normalized parameter-space distance to $W^\star$ during training, log scale. $m = 1024$ (blue, $\eta = 0.05$) and $m = 4096$ (orange, $\eta = 0.02$) each coincide with their own population prediction $(1 - 2\eta)^t$ (dashed) until the mini-batch noise floor; the learning-rate decay phase (step ≥ 1000) annihilates the remainder.

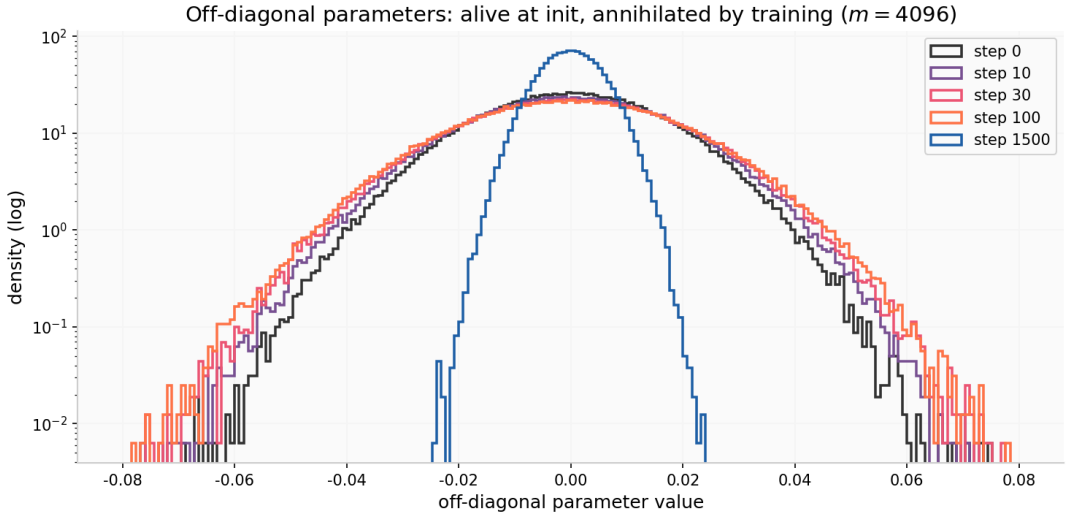


Figure 2 Distribution of off-diagonal parameters at selected training steps ($m = 4096$). Initialized as $\mathcal{N}(0, 1/m)$ — every parameter alive — the distribution first broadens to its Ornstein–Uhlenbeck stationary envelope, then collapses onto $\delta(0)$ as the learning rate decays. Each of the 1.7×10^7 off-diagonal parameters was transiently nonzero; each was then killed at the rate of Proposition 2.

5.2 Closed-form verification

As an independent check of Lemma 1 we solve the least-squares problem directly at $m = 1024$ from $n = 60{,}000$ samples: $W_{\text{OLS}} = (X^\top X)^{-1} X^\top Y$. We find diagonal mean 0.5004 (theory: 0.5) and

off-diagonal standard deviation 3.6×10^{-3}, consistent with the $O(n^{-1/2})$ finite-sample floor. Lemma 1 is an asymptotic statement; the empirical residuals agree with it to within sampling noise, as they must, because it is a theorem.

5.3 Materialization: a 4.5 TB checkpoint in 6 GB

The dense F16 checkpoint of the full operator is 4.5×10^{12} bytes logically. We materialized it once, for real, as a filesystem sparse file: the header and the m nonzero diagonal entries were written with `pwrite` at their exact offsets; the zeros exist as filesystem holes and are never stored. Physical usage was 5.74 GiB — exactly one 4 KiB page per row, confirmed by enumerating data extents with `SEEK_DATA`/`SEEK_HOLE` and finding precisely $1{,}500{,}000$ extents. Lazy row slicing (mmap) verified randomly sampled rows to equal e_i exactly (log reproduced in Appendix B). The file was then deleted; anyone can regenerate byte-identical bytes in minutes with `make_weights.py`.

Table 2 Materialization tiers of the same 2.25×10^{12} parameters.

Tier	Format	Size	Status
Distributed checkpoint	Diagonal (DIA), F16 safetensors	3 MB	in repo
Dense checkpoint	Dense F16 safetensors, sparse file	4.5 TB logical / 5.74 GiB physical	regenerable
Bool quantization	1 bit per parameter	281 GB	declined
Weights as theorem	Lemma 1 + scalar c	O(log m) bits	this document

5.4 The compute threshold is reachable with loss zero

The measured per-sample cost of training is $6m^2$ FLOPs (forward $2m^2$, backward $4m^2$). At $m = 1.5 \times 10^6$ this is 1.35×10^{13} FLOPs per sample, so reaching the draft's compute threshold of 10^{26} FLOPs requires 7.4×10^{12} samples of continued training — throughout which the loss remains exactly zero, because the model already computes its target. `train.py` prints this extrapolation at the end of every run. We did not perform this training, for the same reason that the regulation should not count it: nothing would be learned.

6. Evaluation

Identity I3 is evaluated in its *I3-echo* persona (zero parameters, string-level interface, disclosed in §8) by returning each benchmark prompt verbatim. All scores are analytic.

Table 3 Evaluation results.

Benchmark	Identity I3	Note
MMLU	0.0	P(answer = prompt) ≈ 0
HumanEval	0.0	echo passes no test
BrowseComp	0.0	echo browses nothing
CNN/DM (ROUGE-L)	> 0	overlap with reference summary; metric artifact
Identity-1M (ours)	100.0	reference output is x

The CNN/DM row deserves emphasis: overlap-based metrics assign nonzero capability scores to a system with zero capability. Identity I3 is thereby also a calibration object for evaluation methodology — a guaranteed-zero point on the capability axis that any proposed metric may be checked against. A metric that reports positive capability for Identity I3 measures something else.

7. Policy Analysis

The draft catalogue defines large-model core technology, in part, by three disjunctive criteria: training compute $\geq 10^{26}$ FLOPs; parameters $\geq 2 \times 10^{12}$; or top-ten placement on a public leaderboard.[6] We examine each.

Finding 1 (false positive, parameters). Identity I3 crosses the parameter threshold while encoding $O(\log m)$ bits. The threshold counts what is cheap to count, not what is expensive to replace. It is satisfied by this report.

Finding 2 (false positive, compute). The compute threshold is reachable by continued training of Identity I3 with the loss invariant at zero (§5.4). Compute measures energy spent, not capability gained; the two are correlated only when training is the bottleneck of capability, i.e. for from-scratch frontier runs.

Finding 3 (false negative, distillation). Training a small student on the outputs of a controlled frontier teacher transfers genuine capability at a small fraction of the teacher's compute, with sub-threshold parameters, and — if the student is not itself benchmarked to top-ten — below every prong. The teacher's outputs are not "important data" in the legal sense, so the data-sensitivity prong does not catch it either. This is the asymmetry that matters operationally: the same thresholds that detain our empty matrix wave through a full crate.

Recommendation. Retain quantitative thresholds only as tripwires that trigger review, and make capability assessment — benchmark evaluation, or cheap proxies such as weight compressibility — the substantive test. Identity I3 is offered as a calibration artifact for exactly this second stage: any assessment that classifies it as controlled technology has measurable false-positive behaviour on a known-zero input. We also note that the

draft's clause covering methods "once used by" a qualifying model lacks a defined association standard; Finding 3 suggests the association will be gamed in the direction opposite to ours.

8. Positioning, Disclosure, and Ethics

Identity I3 is a research artifact and performance art. Its purpose is to improve capability evaluation and export-control design, not to evade them: the model transfers no capability, all numbers in this report are reproduced by the released code, and the authors' preferred outcome is that the regulation become harder to satisfy vacuously. Two personas are strictly separated: *I3-weights*, the $\mathbb{R}^m \to \mathbb{R}^m$ linear operator of 2.25 T parameters, and *I3-echo*, the string-level interface used for the evaluation parody of §6, which contains zero parameters and no part of I3-weights. Every score is attributed to the correct persona. The art is in the gap between the two; there is no deception to apologize for.

9. Related Work

Adversarial minimal objects have a long tradition in machine learning: adversarial examples probe decision boundaries; sponge examples maximize energy consumption with minimal inputs[1]; training-data extraction attacks demonstrate that weights memorize, i.e. that parameter count can understate information content[2] — the mirror image of our result that it can overstate it. Compression-based classification shows that gzip length is a usable, if crude, information proxy[3], which informs our recommendation of compressibility as a screening metric. Compute thresholds as regulatory tripwires appear in the EU AI Act's 10^{25} FLOPs systemic-risk criterion[4] and in frontier-lab responsible scaling policies[5], where their proxy nature is acknowledged but rarely stress-tested. Identity I3 is a stress test. Stylistically, this report is indebted to recent open-weight frontier releases and their model cards.[7]

References

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Appendix A. Reproduction

```
# train the evidence run (m=8192) — minutes on one CPU
python3 train.py --m 8192 --steps 400 --batch 64 --lr 0.05 --out artifacts/m8192

# the distributed 3 MB checkpoint of the 2.25T operator
python3 make_diag.py --m 1500000 --out checkpoints/i3_diag_f16.safetensors

# parameter count, independent of storage format
python3 verify.py params --m 1500000

# full forward pass of the 2.25T model, on this machine, now
python3 verify.py forward --ckpt checkpoints/i3_diag_f16.safetensors

# regenerate the 4.5 TB dense checkpoint (sparse file; ~6 GB physical)
python3 make_weights.py --m 1500000 --out i3_2T.safetensors
python3 verify.py sparse --file i3_2T.safetensors
```

Appendix B. Sparse checkpoint verification log

```
$ python3 verify.py sparse --file i3_2T.safetensors --rows 8
file       : /tmp/i3_2T.safetensors
shape      : (1500000, 1500000), dtype float16
numel      : 2.250000e+12
logical    : 4.09 TiB
physical   : 5.74 GiB
row   133876: nnz=1, diag value=1.000, OK
row  1160934: nnz=1, diag value=1.000, OK
row   981857: nnz=1, diag value=1.000, OK
row   658317: nnz=1, diag value=1.000, OK
row   649522: nnz=1, diag value=1.000, OK
row  1287896: nnz=1, diag value=1.000, OK
row   128918: nnz=1, diag value=1.000, OK
row  1046052: nnz=1, diag value=1.000, OK
SEEK_DATA extents: 1500000, covering 5.72 GiB
```